Does a kernel remove long memory from kernel tests?

EXP7, 3 November 2026. All results are simulation and exploratory.

The answer

1. Partly. For testing whether two series are independent, a differenced kernel stops false alarms from rising with memory.
2. It does not fix the extra false alarms that resampling rules make anyway. So our pre-registered size test failed for every kernel.
3. For split MMD, which compares two halves of one series, several memory kernels still climb on heavy-tailed data.
4. Every memory kernel is blind to some dependence that the plain kernel sees, and the blind spots differ.
5. Harris's shift test on the plain kernel already holds false alarms at 1.6% to 3.8% in every stationary case.

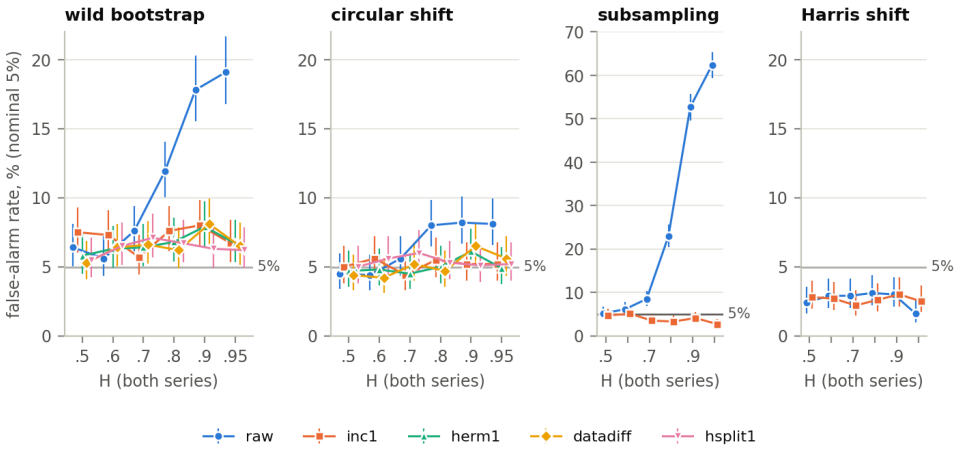
Long memory means correlations fade so slowly that they never add up to a finite total. H measures it: 0.5 means none, and values near 1 mean strong memory.

Does the increment kernel stop false alarms?

What is the increment kernel?

- It compares each point with the one before it, so it looks at changes rather than raw values.
- Changes forget the slow swings that create long memory, so their correlations die out quickly.

It stops the climb with memory. It does not stop the extra false alarms that resampling rules make anyway.



A false alarm is a test reporting dependence between two independent series. The target rate is 5%. In the figures, `inc1` is the increment kernel and `raw` is the plain kernel.

With the plain kernel, false alarms climb as memory grows. At $H = 0.95$, block resampling rejects 19.1% with the plain kernel and 6.7% with `inc1`. Under subsampling the rates are 62.4% and 2.7%.

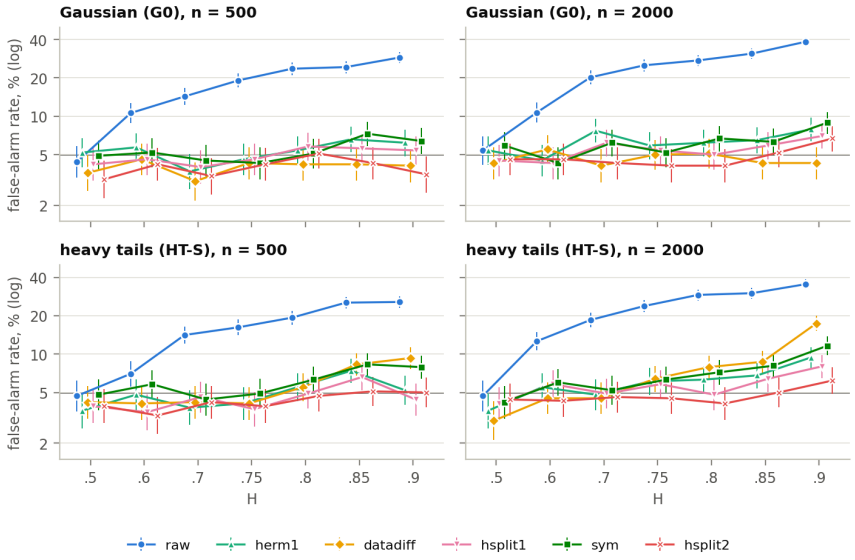
Under block resampling, every memory kernel still rejects 6.4% to 8.3%, even with no memory. So the size test we registered in advance failed for every kernel under the resampling rules. Under the shift rule, which slides one series past the other, they all sit near 5%.

Does it work for split MMD?

Why test with split MMD?

- It asks whether the first half and the second half of one series come from the same distribution.
- That is how you spot a change or regime shift, so a false alarm means a false regime call.

Not reliably. On heavy-tailed data, several memory kernels climb with memory, as the plain kernel does.



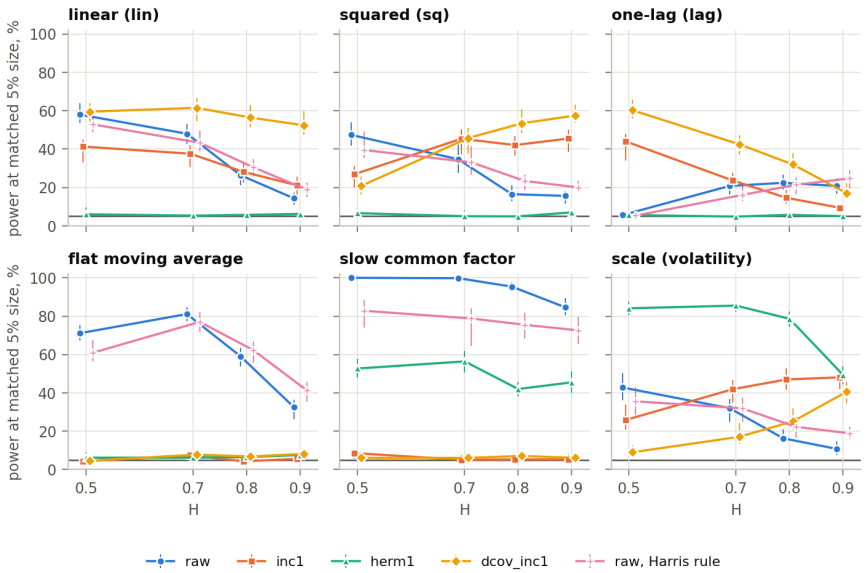
`inc1` cannot be used here, because its sum collapses to three points.

Four kernels climb on heavy tails: `datadiff` (differenced data), `sym` (distance from the median), `herm1` and `hsplit1`. At $H = 0.9$ they reject 17.4%, 11.6%, 9.4% and 8.0%. The plain kernel rejects 35.3%.

`hsplit2` comes closest. Its block rule holds on both data types, and its plug-in rule has one rising cell. That one cell may be chance among many comparisons.

What does it cost?

Each memory kernel is blind to one kind of dependence, and the blind spots differ by kernel.



Power is how often a test finds real dependence. Here each test is first tuned to exactly 5% false alarms. That tuning needs the true null, so it compares statistics, not tests an analyst could run.

`inc1` misses a flat moving average: 4.3% against 71.1% at $H = 0.5$. It also misses a slow common factor: 8.6% against 100.0%. `herm1` misses linear dependence: 6.0% against 58.0%.

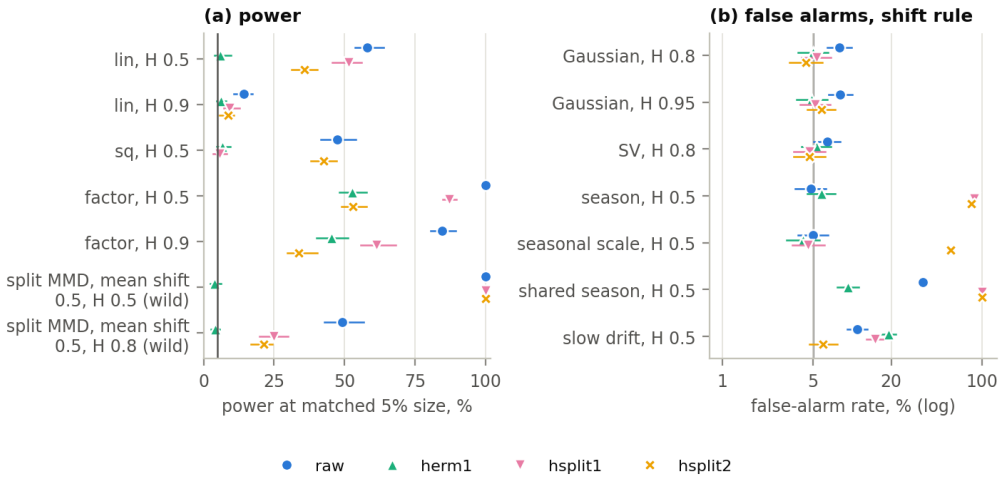
Memory kernels win elsewhere. At $H = 0.9$, `dcov_inc1` (a differenced distance kernel) finds linear dependence 52.4% of the time. The plain kernel finds it 14.3% of the time.

How does the Hermite split do?

What is the Hermite split?

- It removes the low polynomial layers of the normal scores (ranks mapped to a bell curve), where memory sits.
- It then tests those removed layers on their own and combines both results into one decision.

It wins back much of the power that `herm1` loses. It breaks under seasons and has no size proof.



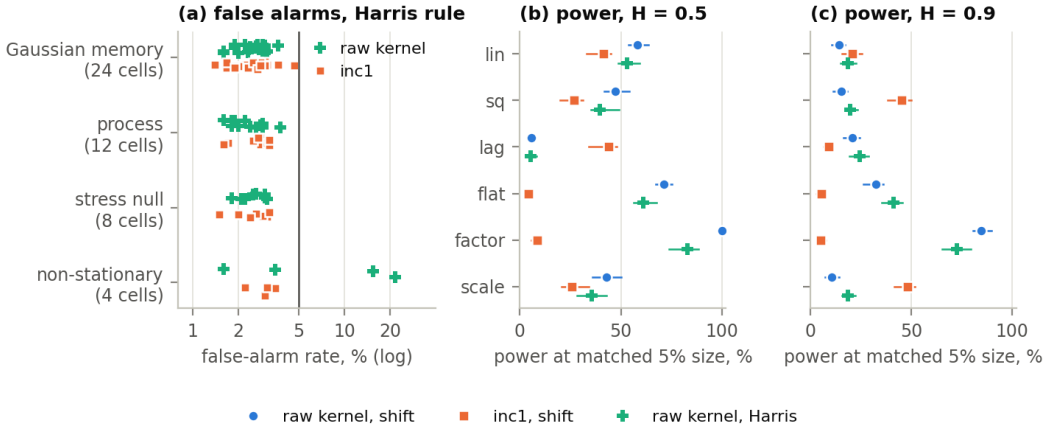
`hsplit1` and `hsplit2` remove one or two layers. `herm1` removes the first layer but skips the separate test.

Linear power at $H = 0.5$ rises from 6.0% with `herm1` to 51.5% with `hsplit1`. The plain kernel has 58.0%. At $H = 0.9$ the gain has gone: `hsplit1` finds 9.2%.

A seasonal pattern with the same period in both series pushes `hsplit1` to 87.0% false alarms, as predicted. Under slowly wandering volatility, the memory sits in layer 2, as the user expected. Yet `hsplit1` stays at 4.7% there.

Is Harris's test enough?

For stationary data it keeps false alarms below 5%, and no memory kernel beats it on every alternative.



Harris's test compares the real alignment of the two series with many shifted alignments. Its false-alarm bound holds at any sample size when the data are stationary. Stationary means their behaviour does not change over time.

It rejects 1.6% to 3.8% in all 44 stationary null cases. A fixed season shared by both series is not stationary. There it rejects 21.5%, while `inc1` rejects 3.0%.

On power it trails the plain kernel at $H = 0.5$, for example 82.8% against 100.0% on the factor. At $H = 0.9$ it is level or ahead. EXP7 checked its exact form only for small shift windows, so its main rates are observed, not proved.

What this means for the paper

- Present `inc1` as a cure for the memory climb in independence tests, not as a fix for size.
- Pair it with the shift rule or Harris's test. Resampling rules over-reject with or without memory.
- Do not claim the cure for split MMD. Only `hsplit2` with its block rule holds there.
- Report levels and increments side by side. Each kernel is blind where the other sees.
- Limit the theory to a fixed kernel width. The data-driven width breaks the proofs at $H \geq 0.9$.

Part 2. Optional: read only if you need the detail

Each section below backs one point in Part 1. Skip any you do not need.

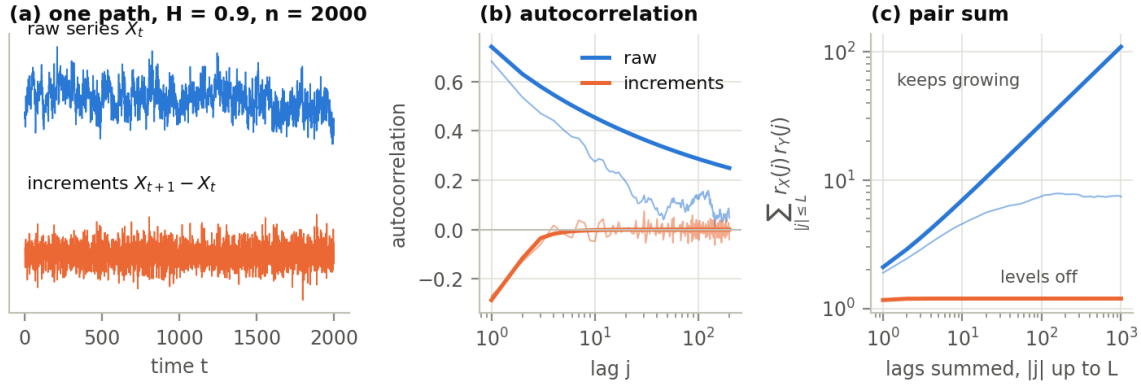
A. Terms

- H **(Hurst exponent).** 0.5 means no memory. Values near 1 mean strong long memory.
- **fGn (fractional Gaussian noise).** The standard simulated long-memory series. All main cells use it.
- **False-alarm rate (size).** How often a test rejects when the two series are truly independent. The target is 5%.
- **Power at matched size.** The hit rate after each test's cut-off is reset to give exactly 5% false alarms. The reset uses the true null, which a real analyst never has.
- **Kernels.** `raw` is the paper's Gaussian kernel. `inc1` differences the kernel features at lag 1. `datadiff` uses a Gaussian kernel on differenced data. `dcov_inc1` differences a distance kernel. `sym` uses distance from the median. `hp10` removes the lowest 10% of frequencies. `herm1` and `herm2` remove the first one or two Hermite layers (polynomial pieces) of the normal scores. `hsplit1` and `hsplit2` add a separate test on the removed layers.
- **Rules** set the cut-off. Wild bootstrap is block resampling with random signs. Subsampling recomputes the statistic on short windows. Circular shift slides one series past the other. Harris shift is a non-circular shift with a finite-sample bound.
- **Wilson interval.** A 95% interval for a rate. With 1,000 replications it is about ± 1.4 points at 5%.
- **Bandwidth.** The width σ of the Gaussian kernel. Every arm sets it to the median distance in the sample.
- **Split MMD.** A two-sample test between the first and second halves of one series.

Scale: 142 cells, 166,400 rows, 0 error rows. 1,000 replications per cell at $n \leq 2000$, 300 at $n = 4000$, 2,000 per growth cell.

B. Why increments forget: one simulated path

The lag-1 differences of a long-memory series have short memory. The sum that drives the null stays finite.



Thick lines: exact fGn theory. Thin lines: this simulated pair (seed 20261103).

Panel (a) shows one fGn path at $H = 0.9$ and its lag-1 differences. The raw path moves in long slow swings. The differences look like plain noise.

The raw autocorrelation is still 0.29 at lag 100 in theory. The differences have -0.29 at lag 1 and are near zero from lag 5 on.

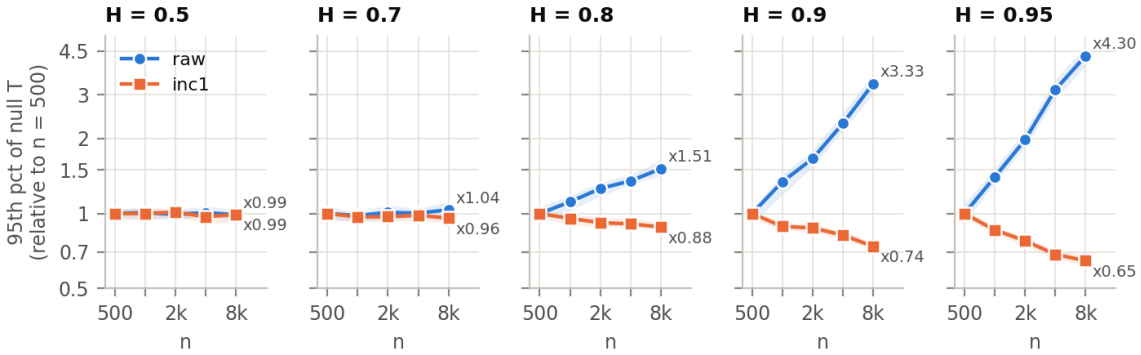
The test's null mean grows with the sum over lags of $r_X(j)\, r_Y(j)$. For the raw pair that sum reaches 109 by lag 1000 and keeps climbing. For differences it stops at 1.19.

THEORY.md proves the same levelling-off for kernel features at every $H < 1$ (Theorem A). That proof assumes a fixed bandwidth and memory at frequency zero only.

> One path hides its own memory. This pair's sample sum is 7.4, against 109 in theory. Subtracting the sample mean biases the sample autocorrelation down. Path: `kmem.legacy.mbcore.fgn`, seed 20261103. Differences: `kmem.memory_kernels.first_difference`.

C. Null growth and the bandwidth effect

The raw null grows with n once H is above 0.75. The `inc1` null falls at high H, and the bandwidth explains most of that fall.



Each panel shows the 95th percentile of T under independence, relative to $n = 500$. Each point uses 2,000 replications. Bands are bootstrap 95% intervals.

At $H = 0.95$, raw grows $\times$4.30 from $n = 500$ to 8000. `inc1` falls to $\times$0.65. With a fixed bandwidth the fall shrinks but stays: $\times$0.84. So the median bandwidth explains most of the fall, but not all of it.

The sample bandwidth is still drifting up at $n = 8000$. Under strong memory the sample spread settles slowly. Putting the mean σ into THEORY's fixed-bandwidth formula matches the EXP7 mean of T within 4% in every cell.

H, n	mean σ	EXP7 mean T_{inc1}	formula at mean σ	THEORY limit ($\sigma = 0.954$)
0.9, 500	0.806	0.336	0.335	0.208
0.9, 8000	0.872	0.265	0.269	0.208
0.95, 500	0.650	0.250	0.239	0.071
0.95, 8000	0.732	0.171	0.168	0.071

- The fall does **not** make a rule conservative. Wild and shift rebuild their reference with the same bandwidth.
- THEORY's Theorems A, B and D assume a fixed bandwidth. At $H \geq 0.9$ they do not describe the computed statistic.
- The raw slope at $H = 0.9$ is 0.425 [0.391, 0.469], against 0.6 in theory. Bandwidth drift may explain part of that gap. Nobody has tested this.
- Only raw and `inc1` ran in the growth grid.

D. False-alarm detail for the independence test

These are the rule-level numbers behind "Does the increment kernel stop false alarms?"

The 24 Gaussian cells are $n = 500$, 2000 and 4000, with eight H pairs. The pre-registered rule needs a valid rate in all 24 cells and no rise in n.

- None of the 16 wild or subsampling pairs passed it (0 of 16).
- "All 24 valid" cannot rank kernels. A rule at exactly 5% fails one cell with probability 2.8%. So it passes all 24 cells with probability only 0.50.
- Pooled over 18,400 decisions, every memory kernel under shift rejects 4.95% to 5.34%. Raw under shift pools at 5.97%.
- Under wild, memory kernels pool at 6.4% to 8.3%. `inc1` pools at 7.70%.
- A logistic trend test in H, with n fixed effects, finds no trend for `inc1`, `herm1`, `herm2`, `sym`, `datadiff` or `hsplit1`. Every p is at least 0.24, under both wild and shift.
- `hp10` under wild climbs: slope +0.62, $p = 0.002$. `hsplit2` under wild falls ($p < 0.001$).
- So prediction P1, "`inc1` holds size at every H", is refuted on the wild half. The shift half (21 of 24 cells) fits exact size.
- Harris's bound covers the literal per-segment form. EXP7 checked that form only at $N = 19$. The main rule uses $N = 99$ at $n = 2000$. So the main Harris rates are observed, not proved.

E. Split MMD detail and the REVIEW.md correction

All split-MMD cells use the wild rule and 1,000 replications.

kernel (heavy tails)	$H = 0.5$, $n = 2000$	$H = 0.9$, $n = 2000$	$H = 0.9$, $n = 500$
`datadiff`	3.0%	17.4%	9.3%
`sym`	4.2%	11.6%	7.9%
`herm1`	3.6%	9.4%	5.2%
`hsplit1`	4.1%	8.0%	4.4%
`hsplit2`	4.4%	6.2%	5.0%

- The rise from $n = 500$ to 2000 at $H = 0.9$ is the memory signature the study tests for.
- `datadiff` holds on Gaussian data (4.3% at $H = 0.9$) but breaks on heavy tails.
- **Correction to REVIEW.md.** Its table marks `hsplit2` on Gaussian data as memory-flat. At $H = 0.9$ the plug-in rule rises from 3.5% [2.5, 4.8] to 6.7% [5.3, 8.4]. The intervals do not overlap. This is one cell among many comparisons, so it is weak evidence.
- The block rule of `hsplit2` stays valid in that cell: 3.6% and 5.4%.
- `herm1` and `herm2` rise on Gaussian data at $H = 0.7$, from $n = 500$ to 2000.
- These flags use the Wilson-overlap rule. Nobody ran a trend test for split MMD.

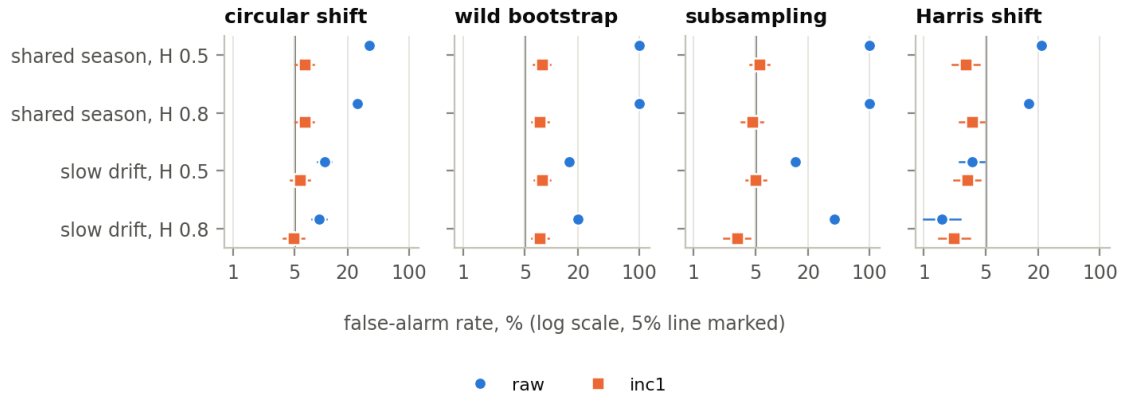
F. Power and Hermite-split detail

- All power cells use $n = 2000$ and the circular-shift rule. Bars are bootstrap 95% intervals. Each alternative has one effect size.
- `inc1` sees one-lag dependence the plain kernel misses: 44.0% against 5.6% at $H = 0.5$.
- `herm1` leads on scale dependence: 84.1% at $H = 0.5$. Raw has 42.8%.
- Raw's linear power falls from 58.0% to 14.3% as H goes from 0.5 to 0.9.
- `hsplit2` recovers squared dependence: 42.6% against `herm1`'s 6.6%. Raw has 47.4%.
- Split MMD mean shift 0.5 at $H = 0.5$: `hsplit1` 100.0%, `herm1` 3.9%. At $H = 0.8$: `hsplit1` 24.9%, raw 49.2%.
- Under shift, `hsplit1` pools at 5.34% and `hsplit2` at 5.17% over the 24 Gaussian cells.
- `hsplit2` rejects 57.5% with a seasonal scale. The low-layer test has no shift reference. This was predicted (P7).
- Under slow drift, `hsplit1` rejects 15.0%. The remainder causes this, not the low layer. `herm1` alone rejects 19.2%.
- Stochastic volatility (SV), $H = 0.8$: estimated memory is 0.518 in layer 1 and 0.892 in layer 2. `hsplit1` rejects 4.7% under shift and 7.2% under wild. `hsplit2` rejects 4.7% and 4.7%. Raw shift rejects only 6.4%, so this cell is a weak test.
- THEORY.md finds that `hsplit2` leaves 2.87% of the rank-one memory. So any cure under SV is at finite n only.
- Plug-in against true layer memory moved `hsplit` size by at most 0.4 points in 54 comparisons.

> Matched size uses the true null of each simulated cell. An analyst cannot do this. A different effect size could reorder the kernels.

G. Seasonal and process cells

Differencing removes the false alarms from a shared fixed season and from slow drift. The plain kernel fails there under every rule.



"Shared season" gives both series a seasonal mean with the same fixed phase. "Slow drift" gives both a scale that drifts with period $4n$. Both are non-stationary on purpose, with $n = 2000$.

- Shared season, $H = 0.5$: raw rejects 35.2% under shift. `inc1` rejects 6.5%.
- Slow drift, $H = 0.5$: raw rejects 11.0% under shift. `inc1` rejects 5.8%.
- Kernels that keep the level carry the season. At slow drift, `herm1` rejects 19.2% and `sym` 15.6%.
- `hsplit1` rejects 100.0% in the shared season. RESULTS.md first omitted this cell.
- Harris's theorem needs stationarity, so it does not cover these cells. `inc1`'s low Harris rates here are observed only.

H. Theory status

"Proved" means proved in THEORY.md or a source it opened.

Proved

1. The increment kernel's null mean stays bounded at every $H < 1$, for fGn-type memory and a fixed bandwidth (Theorem A).
2. The increment kernel is exactly blind to flat three-lag dependence. On split MMD it collapses to three points.
3. Harris's shift bound holds at finite n for any statistic, when the shifted series is stationary.
4. No location-invariant test detects a level shift below $\delta \sim n^{H-1}$ in the layer that carries the memory.

Rests on a preprint

- An argument shows the shift rule's size is right asymptotically (`inc1`, and raw below 3/2), but it rests on an unreviewed preprint.

Only simulated (exploratory)

1. Every rate in this report: 142 cells, one setting per process, one effect size per alternative.
2. The size of `hsplit1` and `hsplit2`. No part of either test has a size proof.
3. Every arm uses a median bandwidth, while the proofs assume a fixed one. At $H \geq 0.9$ the proofs do not describe the computed statistic.
4. `dcov_inc1`, `sym` and `hp10` have no exact theory. Wild multipliers on increments have no theory.
5. The main Harris columns use whole-series bandwidths. The literal form was checked at $N = 19$ only.

Failed or refuted (kept on purpose)

1. The pre-registered size rule failed under wild and subsampling for every kernel (0 of 16 pairs). P1 (`inc1`) and P2 (`datadiff`) are refuted on the wild half. They pool at 7.70% and 7.18%.
2. "Memory kernels stop the climb" fails for split MMD. On heavy tails `datadiff`, `sym`, `herm1` and `hsplit1` climb with H. `hp10` climbs under wild in the independence test too.
3. The median bandwidth explains most of the falling `inc1` null ($\times$0.65, against $\times$0.84 at a fixed bandwidth). RESULTS.md first called it conservative, and that reading does not hold.
4. The layer-2 memory unit check read 0.707 against 0.8. The threshold was relaxed after the failure was seen.
5. Other refuted expectations: `herm1` and `sym` failing at $H \geq 0.9$, and Harris losing power under memory. Three theory claims also fell: a bounded null for any process, "deflation removes nothing under SV", and "the oracle filter sums to 1".

I. What to test next

1. Harris on each Hermite layer plus Bonferroni. This is the one `hsplit` version with a size proof. EXP7 did not run it.
2. Rerun growth and size with a fixed bandwidth, to separate the kernel from the bandwidth drift.
3. Split MMD on heavy tails at larger n, read with a trend test in H.
4. `hsplit2` on SV at $n \geq 16{,}000$, where theory says the memory effect still grows.
5. A combined test that reports levels and increments together, scored on the flat and factor alternatives.